Received January 11, 2025; | Revised April 6, 2025; | Accepted April 6, 2025

DOI: xxx/xxxx

RESEARCH ARTICLE

Principal Component-Based Classification of Blazar Candidates of Uncertain Type in the 4FGL/LAT

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We adopted Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and Support Vector Machines (SVM), to study classification for a sample of 2708 Fermi blazars: 1142 BL Lac objects (BLLs), 759 flat spectrum radio quasars (FSRQs), and 807 blazar candidates of uncertain type (BCUs). From nine observables, the first three PCA components explained 78.4% of the variance, with the main differences among blazars linked to spectral shape, luminosity, and synchrotron dominance. We introduced a dimensionless separability metric to quantify and compare the ability of different observable to distinguish BLLs from FSRQs. Using known BLLs and FSRQs, we trained an LDA model that achieved 90.6% accuracy ($\kappa = 0.81$) and yielded 339 BLLs and 468 as FSRQs candidates for the 807 BCUs, while the SVM model achieved 83% accuracy yielding 349 BLL and 384 FSRQ candidates, with 71 identified as low-confidence cases. The two models agreed on 81.7% of the BCUs with most differences occurring near classification boundaries. Combining LDA and SVM improves classification reliability: LDA reveals the global separation pattern, while SVM supports the linearity of LDA. These results indicate that the BLL-FSRQ distinction reflects genuine physical differences and provide a robust approach for identifying uncertain sources.

KEYWORDS:
galaxies: active, BL Lacertae objects: general, quasars: general, galaxies: jets

1 | INTRODUCTION

Blazars are a subclass of active galactic nuclei (AGNs) characterized by relativistic jets closely aligned with the line of sight, resulting in extreme observational properties such as strong variability, high luminosities, and non-thermal emission across the electromagnetic spectrum (Ballet, Bruel, Burnett, Lott, & Fermi-LAT Collaboration,

[0]**Abbreviations:** PCA, Principal Component Analysis; LDA, Linear Discriminant Analysis; SVM, Support Vector Machine; BL Lac, BL Lacertae object; FSRQ, Flat Spectrum Radio Quasar; BCU, Blazar Candidate of Uncertain type; AGN, Active Galactic Nucleus; SED, Spectral Energy Distribution; CD, Compton Dominance; EC, External Compton.

2023; G. Chen et al., 2026; Fan, 2003; Fan et al., 2021; Stickel, Padovani, Urry, Fried, & Kuehr, 1991; Urry & Padovani, 1995; Wills, Wills, Breger, Antonucci, & Barvainis, 1992).

Based on their optical emission-line properties, blazars are traditionally divided into two subclasses: BL Lac objects (BL Lacs) and flat-spectrum radio quasars (FSRQs). BL Lacs exhibit either very weak or no emission lines ($EW < 5\,\mathrm{\AA}$), whereas FSRQs show strong emission lines ($EW > 5\,\mathrm{\AA}$). Various quantitative criteria have been proposed for classifying these subclasses. For example, Ghisellini, Tavecchio, Foschini, and Ghirlanda (2011) suggested using the ratio of broad-line region (BLR) luminosity ($L_{\rm BLR}$) to the Eddington luminosity ($L_{\rm Edd}$), with $\frac{L_{\rm BLR}}{L_{\rm Edd}} \sim 5 \times 10^{-4}$ as the dividing line between the two classes. Similarly, Pei, Fan, Yang, Huang, and Li (2022) defined an "appealing zone" for classification with $2 \times 10^{-4} \lesssim \frac{L_{\rm BLR}}{L_{\rm Edd}} \lesssim 8.51 \times 10^{-3}$, whereas H. Xiao et al. (2022) proposed a relation incorporating γ-ray luminosity, $\log \frac{L_{\rm BLR}}{L_{\rm Edd}} = 0.25 \log \frac{L_{\gamma}}{L_{\rm Edd}} - 2.23$, to separate BL Lacs from FSRQs.

Those studies demonstrate that although emission-line properties provide a useful basis for blazar classification, spectroscopic observations need a lot of observation time, and the the boundaries between BL Lacs and FSRQs are not uniquely defined and often vary across different schemes. This variation reflects the continuous and multi-dimensional nature of blazar properties, highlighting the limitations of classifications based on a single observable.

Indeed, most existing schemes rely on only one or two observational indicators, such as emission-line strength, synchrotron peak frequency, or γ-ray spectral properties. Although these approaches can separate subclasses to some extent, it remains unclear whether a small number of observables are sufficient to capture the full diversity of blazar properties. Moreover, the optimal combination of parameters for distinguishing BL Lacs and FSRQs has not been established, and the number of independent components driving the differences between those subclasses is still unknown.

These uncertainties are particularly critical for blazar candidates of uncertain type (BCUs), which often exhibit intermediate or ambiguous properties that cannot be reliably classified using existing low-dimensional criteria. Consequently, a systematic multivariate analysis is needed to identify the dominant components shaping blazar diversity and to explore how different combinations of observational and physical parameters contribute to the separation of blazar subclasses. Such an approach can provide a more robust framework for classification and deepen our understanding of the intrinsic differences among blazar populations.

Early studies have shown that the synchrotron peak frequency of blazars provides an important observational dimension for classification. Padovani and Giommi (1995) calculated the spectral energy distributions (SEDs) of a sample of BL Lac objects, including X-ray-selected BL Lacs (XBLs) and radio-selected BL Lacs (RBLs), and found that XBLs typically exhibit higher synchrotron peak frequencies, with $\log \nu_{\rm p}(\mathrm{Hz}) > 15$, whereas most RBLs have $\log \nu_{\rm p}(\mathrm{Hz}) \leq 15$. Based on this distinction, they proposed the subclasses of high-peak-frequency BL Lacs (HBLs) and low-peak-frequency BL Lacs (LBLs).

Following the classification scheme introduced by Abdo et al. (2010), Fan et al. (2016) derived SEDs for a large sample of 1392 Fermi blazars using a log-parabolic model, $\log \nu f_{\nu} = p_1(\nu - p_2)^2 + p_3$, and proposed dividing blazars into low synchrotron peak sources (LSPs), intermediate synchrotron peak sources (ISPs), and high synchrotron peak sources (HSPs), with boundaries at $\log \nu_{\rm p}(\mathrm{Hz}) = 14.0$ and 15.3. More recently, J. H. Yang et al. (2022) calculated SEDs for 2709 blazars in the 4FGL-DR3 catalogue and suggested slightly different boundaries at $\log \nu_{\rm p}(\mathrm{Hz}) = 13.7$ and 14.9. Despite those differences, the studies consistently support the existence of three broad synchrotron-based subclasses, separated approximately at $\log \nu_{\rm p}(\mathrm{Hz}) \sim 14$ and ~ 15, namely LSPs, ISPs, and HSPs (Fan et al., 2025).

In the γ-ray band, blazars exhibit extremely high luminosities and distinct spectral behaviors. Ghisellini, Maraschi, and Tavecchio (2009) reported a positive correlation between the γ-ray photon spectral index and γ-ray luminosity, with FSRQs preferentially occupying the region of higher luminosity and softer spectra, while BL Lacs tend to show lower luminosities and harder spectra. However, Fan et al. (2025) argued that this correlation is apparent rather than intrinsic, as an anti-correlation emerges when LSPs, ISPs, and HSPs are considered separately, as found by Yang et al. (2022).

Several empirical boundaries in the γ-ray parameter space have been proposed to separate BL Lacs and FSRQs. For example, L. Chen (2018) suggested using the relation $\Gamma_{\gamma} = -0.127 \log L_{\gamma} + 8.18$ as a dividing line in the Γ_{γ}–L_{γ} plane. Using a support vector machine (SVM) approach, Yang et al. (2022) further demonstrated that BL Lacs and FSRQs can be separated by $\Gamma_{\gamma} = 0.048 \log L_{\gamma} + 4.498$ in the Γ_{γ}–L_{γ} plane, and by $\log V.I. = -10.119\Gamma_{\gamma} + 24.855$ in the variability index versus photon spectral index plane. These relations have been widely applied to classify BCUs, as discussed in previous studies (Agarwal, 2023; Fan et al., 2022; Kang et al., 2019; Kang, Lyu, Wu, Zheng, & Fan, 2024; Kang, Zheng, & Wu, 2023; P.-Y. Xiao et al., 2023).

The detection of γ-ray emissions from a large number of AGNs by Fermi/LAT represents a major observational breakthrough (Abdo et al., 2010; Abdollahi et al., 2022; Ajello et al., 2020), and γ-ray emission was recognized as a defining observational property of blazars (Fan, Yang, Zhang, Li, & Liu, 2013). According to the latest Fermi/LAT catalogues, among 4219 detected AGNs, 3993 are blazars, including 840 FSRQs, 1495 BL Lacs, and 1658 BCUs, indicating that approximately 94.7% of Fermi-detected AGNs are blazars and that BCUs account for about 41.5% of the blazar population (Fan et al., 2025). The large fraction of BCUs highlights the limitations of existing low-dimensional classification criteria.

The physical origin of γ-ray emissions in blazars remains an open question. In leptonic models, γ-rays are produced via inverse Compton (IC) scattering by relativistic electrons in the jet, where the seed photons can originate from synchrotron radiation itself (SSC; Massaro, Perri, Giommi, & Nesci, 2004), or from external photon fields such as the accretion disk (Dermer & Schlickeiser, 1993), the broad-line region (BLR; Sikora, Begelman, & Rees, 1994), or the dust torus (DT; Błażejowski, Sikora, Moderski, & Madejski, 2000). Given the large number of BCUs, a reliable statistical classification of BL Lacs and FSRQs is essential for advancing our understanding of the blazar emission mechanisms.

SED modeling of both the synchrotron and inverse Compton components (Fan et al., 2016; J. Yang et al., 2023; J. H. Yang et al., 2022) shows that FSRQs generally exhibit higher γ-ray luminosities ($\log L_{\gamma}$), higher inverse Compton luminosities ($\log L_{\rm IC}$), higher synchrotron peak luminosities ($\log L_{\rm Syn}$), and lower synchrotron peak frequencies than BL Lacs. In addition, FSRQs typically show a higher Compton dominance, defined as $CD = \log(L_{\rm IC}/L_{\rm Syn})$ (Yang et al., 2022). These systematic but overlapping differences further suggest that no single parameter can fully capture the diversity of blazar properties.

Motivated by those considerations, we adopted a combined multivariate approach based on Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), and Support Vector Machines (SVM). We first applied PCA to identify the dominant components governing blazar diversity and to explore subclass distributions in the reduced parameter space (Mao et al., 2016; Xu, Huang, Deng, Mei, & Wang, 2020). We then trained LDA on known BL Lacs and FSRQs to determine the optimal discriminant axis (Mao et al., 2016), while employing SVM as an independent classifier to validate and cross-check the separations identified by LDA (Fan et al., 2022; Kang et al., 2019; Zhu et al., 2023). We subsequently applied those methods to the BCU population to assign probable subclasses. This paper is organized as follows: Section 2 describes the sample; Section 3 presents the methodology and results; Section 4 discusses the implications; and Section 5 summarizes the conclusions.

2 | SAMPLE

The spectral energy distribution (SED) of quasars exhibits a characteristic two-bump structure, extending from radio to high-energy bands, which is one of their most prominent features (Böttcher, 2007; Fossati, Maraschi, Ghisellini, & Celotti, 1998; Ghisellini, Celotti, Fossati, Maraschi, & Comastri, 1998). To conduct a systematic statistical study of the SED distributions of blazars, it is necessary to construct a large multi-parametric sample that captures their key physical properties.

We adopted the latest catalog of active galactic nuclei released by *Fermi*/LAT, which contains 3743 quasars and BCUs (Abdollahi et al., 2022; Tuo et al., 2024; J. Yang et al., 2023). We imposed no additional selection criteria beyond the availability of SED-related parameters, retaining all sources satisfying these conditions to minimize potential selection biases introduced by predefined subclass assumptions. Redshifts z are available for 2397 sources, ranging from 0.01 to 4.31, with an overall average of $\langle z \rangle = 0.854 \pm 0.671$. Specifically, BL Lacs span $z = 0.01$ to 4.00 with a mean of 0.531 ± 0.501; FSRQs range from 0.08 to 4.31 with a mean of 1.201 ± 0.647; BCUs span 0.01 to 3.86, with a mean of 0.916 ± 0.721. We further discuss the possible impact of redshift-related effects on luminosity-based analyses in Section 4.

Among those sources, the peak frequencies, peak luminosities, and spectral curvatures of the synchrotron and inverse Compton components have been measured for 2708 sources (J. Yang et al., 2023; J. H. Yang et al., 2022). For those sources (1142 BL Lacs, 759 FSRQs, and 807 BCUs), we constructed a multidimensional sample characterized by 13 observational parameters for subsequent analysis. Those parameters include the peak frequencies and luminosities of the synchrotron and inverse Compton components, spectral curvatures, the frequency separation $\Delta\nu = \log\nu_{\rm IC} - \log\nu_{syn}$, the Compton dominance parameter $CD = \log(L_{\rm IC}/L_{syn})$, and the photon spectral index Γ_{PL} in the high-energy band (Abdo et al., 2010; Fan et al., 2016), which together describe both the shape and the radiative output of the SED.

As a descriptive characterization of the sample, we note that the peak luminosities of the synchrotron and inverse Compton components exhibit a strong positive correlation. For the full sample, the relation can be expressed as

$$\log L_{\rm IC} = (1.09 \pm 0.01) \log L_{syn} - 4.08 \pm 0.01, \quad (1)$$

with a Pearson correlation coefficient of $r = 0.930$. The corresponding relations for individual subclasses are:

TABLE 1 Luminosity statistics for blazars.

Band	Class	N	Range	Average
z	Blazar	1594	0.00~4.31	0.85±0.67
z	BL Lac	844	0.00~4.00	0.53±0.50
z	FSRQ	748	0.08~4.31	1.20±0.65
z	BCU	198	0.01~3.86	0.90±0.72
log(CD)	Blazar	1901	-2.00~2.40	-0.03±0.62
log(CD)	BL Lac	1142	-2.00~1.81	-0.32±0.47
log(CD)	FSRQ	759	-3.34~2.22	0.21±0.63
log(CD)	BCU	807	-3.34~2.22	0.21±0.64
$\log L_{\rm p}$	Blazar	1901	36.47~47.74	45.35±0.92
$\log L_{\rm p}$	BL Lac	1142	34.67~47.61	45.04±0.93
$\log L_{\rm p}$	FSRQ	759	42.89~47.74	45.83±0.67
$\log L_{\rm p}$	BCU	807	39.74~47.79	45.29±0.64
$\log L_{\rm R}$	Blazar	1641	31.66~45.39	42.23±1.24
$\log L_{\rm R}$	BL Lac	966	31.66~44.09	41.53±1.03
$\log L_{\rm R}$	FSRQ	675	39.94~45.33	43.24±0.74
$\log L_{\rm R}$	BCU	624	37.23~44.79	42.33±0.74
$\log L_{\rm O}$	Blazar	1215	34.87~47.38	45.10±0.85
$\log L_{\rm O}$	BL Lac	883	34.87~47.29	44.92±0.83
$\log L_{\rm O}$	FSRQ	332	43.86~47.38	45.57±0.69
$\log L_{\rm O}$	BCU	227	42.63~47.01	45.45±0.63
$\log L_{\rm X}$	Blazar	1119	40.61~47.72	44.82±0.93
$\log L_{\rm X}$	BL Lac	670	40.61~47.72	44.66±1.00
$\log L_{\rm X}$	FSRQ	377	41.75~47.48	45.06±0.75
$\log L_{\rm X}$	BCU	72	39.97~46.76	44.62±1.25
$\log L_{\gamma}$	Blazar	1901	34.50~48.79	45.12±1.22
$\log L_{\gamma}$	BL Lac	1143	34.50~47.54	44.57±1.07
$\log L_{\gamma}$	FSRQ	759	42.71~48.79	45.96±0.91
$\log L_{\gamma}$	BCU	807	38.87~47.52	45.28±0.72

- BL Lacs: $\log L_{IC} = (1.10 \pm 0.03) \log L_{syn} - 4.01 \pm 0.07$, $r = 0.809$;
- FSRQs: $\log L_{\rm IC} = (0.76 \pm 0.03) \log L_{syn} + 11.05 \pm 0.09$, $r = 0.617$;
- BCUs: $\log L_{\rm IC} = (0.99 \pm 0.01) \log L_{syn} - 0.26 \pm 0.021$, $r = 0.899$.

FSRQs tend to occupy regions of higher peak luminosities than BL Lacs in these distributions, suggesting that the joint parameter space may contain useful information for subclass separation.

Based on this sample, we employed Principal Component Analysis (PCA) to reduce the dimensionality of the parameter space, followed by Linear Discriminant Analysis (LDA; Jolliffe, 2002) to determine the optimal axis separating BL Lacs and FSRQs. We further introduced Support Vector Machines (SVMs) as an independent classification tool to complement LDA in cases of non-linear separability (Fan et al., 2022; Kang et al., 2019). In this framework, we used BL Lacs and FSRQs with established classifications to define the discriminant boundaries, while including BCUs to probe their distribution in the same parameter space and to assign probabilistic subclass labels without influencing the construction of the class boundaries.

Detailed information and classification results for individual sources are summarized in Table 4 . The full table is provided in a machine-readable format in the appendix.

3 | METHODOLOGY AND RESULTS

3.1 | Methodology

From our multi-dimensional blazar sample, we further extracted nine parameters that are closely linked to the intrinsic physical properties of the sources. We then applied PCA to explore the structural features and dominant patterns of variation in high-dimensional parameter space. PCA has been applied to study blazar variability and classification (Gallant, Gallo, & Parker, 2018; Mao et al., 2016; Xu et al., 2020). As an unsupervised dimensionality reduction technique, PCA identifies directions of maximum variance, enabling us to uncover key physical drivers and to examine the global distribution of sources. This analysis provides initial insight into whether BL Lacs and FSRQs are distinguishable based on their physical properties. We emphasize that these methods are not used as independent or competing classifiers, but as complementary tools addressing different aspects of the problem: PCA is employed to reveal the intrinsic multivariate structure of the parameter space without class labels, LDA provides an interpretable supervised projection optimized for class separation, and SVM serves as an independent validation of the robustness of the classification boundaries. This hierarchical design allows us to separate physical interpretation from classification performance and avoids over-reliance on any single method.

To further investigate this separation and perform classification, we subsequently employed Linear Discriminant Analysis (LDA), a supervised method that finds the linear combination of features that best separates predefined classes (Mao et al., 2016). LDA not only offers a clear projection axis (LD1) for visualizing group separation, but also facilitates the probabilistic classification of sources with BCUs. Its interpretability and statistical foundation make it a valuable tool for understanding the discriminative structure in the parameter space.

Building on the results of LDA, we further applied Support Vector Machines (SVM) to refine and validate the classification. SVM is particularly effective in handling high-dimensional data and capturing complex, nonlinear boundaries through the use of kernel functions (Fan et al., 2022; Kang et al., 2019; Zhu et al., 2023). While LDA provides an interpretable linear boundary, SVM serves as a complementary method capable of modeling more intricate decision surfaces. We trained the SVM on labeled sources and applied the model to classify the BCUs, allowing us to assess the consistency between different classifiers and gain deeper insight into ambiguous or transitional cases.

3.1.1 | Separability Metric

Previous studies and our preliminary analysis suggest that the two source classes under investigation exhibit varying degrees of separation along different parameter dimensions. To systematically quantify this separation, we introduce a general metric, denoted as R, which can be consistently applied throughout our analysis pipeline.

The separability metric R is defined as:

$$R = \frac{W_d}{\sigma_{\rm hm}}, \tag{2}$$

where W_d represents the first-order Wasserstein distance (also known as the Earth Mover's Distance, EMD) between the distributions of two classes, and $\sigma_{\rm hm}$ is the harmonic mean of their standard deviations:

$$\sigma_{\rm hm} = \frac{2\sigma_1\sigma_2}{\sigma_1 + \sigma_2}. \tag{3}$$

We note that the definition of R does not introduce any tunable parameters and does not depend on the classification model itself. It is purely distribution-based and can therefore be applied consistently to both raw observational parameters and derived projection spaces (e.g., PCA and LDA), allowing for a fair comparison of separability across different representations. This formulation incorporates both the inter-class distance and the intra-class spread, offering a robust and interpretable measure of separability.

Formally, the Wasserstein distance between two one-dimensional probability distributions P and Q is given by:

$$W_d(P, Q) = \int_{-\infty}^{\infty} \left| F_P(x) - F_Q(x) \right| \, \mathrm{d}x, \tag{4}$$

where $F_P(x)$ and $F_Q(x)$ are their respective cumulative distribution functions (CDFs). The Wasserstein distance remains finite, continuous, and physically meaningful even when the supports of the two distributions do not overlap (Peyré & Cuturi, 2019; Rubner, Tomasi, & Guibas, 2000; Villani, 2009), making it particularly suitable for astronomical datasets with heavy-tailed or shifted distributions.

The index R thus serves as a quantitative measure of how well two classes are separated in any given parameter dimension: larger R values indicate stronger separability and greater discriminative power.

In the following subsections, we will first apply R to the principal component space (Section 3.1.2), then assess its behavior in the optimal discriminant space given by LDA (Section 3.1.3), and finally validate the classification performance using SVM (Section 3.1.4).

3.1.2 | Principal Component Analysis (PCA)

In this framework, PCA serves to reduce dimensionality and suppress noise (Abdi & Williams, 2010; Jolliffe, 2002), while SVM complements the analysis by modeling the classification of BCUs (Cortes & Vapnik, 1995a; Fisher, 1936). Together, these methods provide a clearer understanding of the structure of blazar populations and offer robust, multi-model predictions of BCU classifications (Doert & Errando, 2014; F. Massaro et al., 2013), improving both the reliability and interpretability of the results.

To ensure that PCA reflects intrinsic physical properties rather than redshift-driven observational effects, we excluded the synchrotron peak flux (P_3) and the high-energy peak flux (P_6), both of which show strong correlations with z (Meyer, Fossati, Georganopoulos, & Lister, 2011). This exclusion is motivated by the goal of isolating intrinsic spectral properties rather than observational distance effects. We verified that including these flux-related parameters does not qualitatively alter the dominant PCA structure, but significantly increases redshift-driven variance, which obscures the physical interpretation of the principal components. The subsequent analysis was based on nine carefully selected parameters that more accurately characterize the physical nature of blazars: synchrotron peak curvature (P_1), synchrotron peak frequency (P_2), synchrotron peak frequency luminosity ($\log L_P$), high-energy peak curvature (P_4), high-energy peak frequency (P_5), gamma-ray luminosity ($\log L_\gamma$), peak frequency difference between the high-energy and synchrotron components ($\Delta\nu$), Compton dominance (CD), and high-energy photon index ($\Gamma_{\rm PL}$). These parameters are complete across the entire sample, providing a reliable basis for the PCA.

To further investigate the structural differences between BLLs and FSRQs in parameter space, we performed PCA separately on the two subclasses. Performing PCA separately for the two subclasses is not intended to redefine the classification, but to assess whether their internal parameter correlations share a common structure. Differences in the resulting component dimensionality and orientations therefore reflect intrinsic complexity rather than imposed separation. The same set of variables was used to ensure consistency in the projection and interpretation.

3.1.3 | Linear Discriminant Analysis (LDA)

PCA reduces the blazar sample to a low-dimensional parameter space, where BLL and FSRQ sources exhibit notable separability along PC_1. To further identify the direction that best discriminates the two classes, we introduced LDA, a supervised learning method that maximizes the ratio of between-class variance to within-class variance (Fisher, 1936; McLachlan, 1992).

We trained the LDA model on sources with known types (BLL and FSRQ) to determine the optimal discriminant axis,

LD_1, which defines the direction that best separates the two classes in the high-dimensional feature space.

The resulting discriminant function is given by:

$$\begin{aligned} LD_1 = {} & 6.51\, P_1 - 2.83 P_4 + 1.27 \Gamma_{PL} \\ & - 0.48 P_2 - 0.47 P_5 + 0.29 \log L_p \\ & + 0.17 \log L_\gamma - 0.14 CD + 0.04 \Delta\nu. \end{aligned} \tag{5}$$

Each source can thus be projected onto this axis and assigned a scalar score that reflects its position relative to the two classes.

3.1.4 | Support Vector Machine (SVM)

SVM is a versatile and widely used machine learning (ML) algorithm, particularly effective in classification tasks. Originating from the work of Cortes and Vapnik (1995b), SVM operates on the principle of finding a hyperplane that best separates data points into different classes in a feature space. The strength of SVM lies in its ability to handle high-dimensional data and perform well with a clear margin of separation, making it robust against overfitting in many scenarios (Burges, 1998).

In `Python`, SVM functionalities are extensively supported by the `scikit-learn` library, a powerful tool for ML that provides simple and efficient tools for data analysis and modeling (Pedregosa et al., 2011). Among its various implementations of SVM, `sklearn.svm.LinearSVC` is a class that specifically handles the linear variant of the algorithm. As the name suggests, `LinearSVC` (Support Vector Classification) applies a linear kernel to perform classification tasks. This method is particularly useful when dealing with large numbers of features, as it scales relatively well and is more computationally efficient compared to SVM with non-linear kernels (Pedregosa et al., 2011).

`LinearSVC` in `scikit-learn` is particularly adept at handling sparse data and has proven effective in text classification problems, where high dimensionality is a common challenge (Joachims, 1998). Users can also customize the regularization parameter `C` to control the trade-off between achieving a low training error and a low testing error, which is crucial in preventing overfitting (Franklin, 2005). The strength of the regularization is inversely proportional to `C`.

3.2 | Results

3.2.1 | Multivariate Structure Analysis Results (PCA)

PCA was applied to nine observed parameters, yielding nine orthogonal components (PC_1 through PC_9). As shown in Figure 1 , the first five components explain 47.7%, 16.9%, 13.7%, 10.7%, and 8.5%, respectively, each showing a certain level of significance and cumulatively accounting for 97.6% of the total variability. However, to ensure model simplicity while preserving the primary structural information, this study adopts the first three principal components (which together explain 78.4% of the variance) as the basis for subsequent analyses. This choice provides a good balance between information retention and efficiency in visualization and modeling.

The second part of the PCA analysis explores the relationships between the principal components and the original variables. Table 2 presents the loadings of the nine variables on the three strong principal components, which can be interpreted as the correlation coefficients between the parameters and the components. Figure 2 A shows the variable loadings on the $PC_1 - PC_2$ plane, while Figure 2 B displays those on the $PC_1 - PC_3$ plane, illustrating the direction and contribution of each parameter. Figures 2 C, D, and E show the contribution rates of each variable to PC_1, PC_2, and PC_3, respectively.

The results show that PC_1 is primarily dominated by the synchrotron peak frequency (P_2), the high-energy peak frequency (P_5), and the high-energy photon index ($\Gamma_{\rm PL}$). An increase in PC_1 corresponds to a decrease in both the synchrotron and high-energy peak frequencies, a softening of the high-energy photon spectrum, an enhancement in high-energy luminosity and Compton Dominance (CD), and an increase in the curvature of the synchrotron peak. This principal component likely reflects the overall spectral shape and energy distribution from radio to optical and high-energy bands.

PC_2 is significantly correlated with the synchrotron peak luminosity ($\log L_P$), the frequency difference between the high- and low-energy peaks ($\Delta\nu$), and the high-energy luminosity ($\log L_\gamma$). As PC_2 increases, the source's total radiative luminosity strengthens, and the offset between the high-energy and synchrotron peak frequencies becomes larger. This component may represent the co-evolution of radiation intensity and frequency structure, capturing changes in the overall energy output.

PC_3 is mainly influenced by the synchrotron peak luminosity ($\log L_P$), the peak frequency difference ($\Delta\nu$, with a negative loading), and the synchrotron peak curvature (P_1). A higher PC_3 value indicates a more prominent synchrotron component (increased luminosity), with a more concentrated spectral shape (decreasing $\Delta\nu$ and increasing P_1). Thus, this component may reveal the increasing dominance of synchrotron radiation in the overall spectral energy distribution and trends toward more focused multi-band emission.

In summary, PCA effectively reduces the complexity of the nine-dimensional physical parameter space into three principal components with clear physical meanings. This simplification aids in characterizing the dominant variation patterns in quasar sources and supports subsequent classification and modeling efforts.

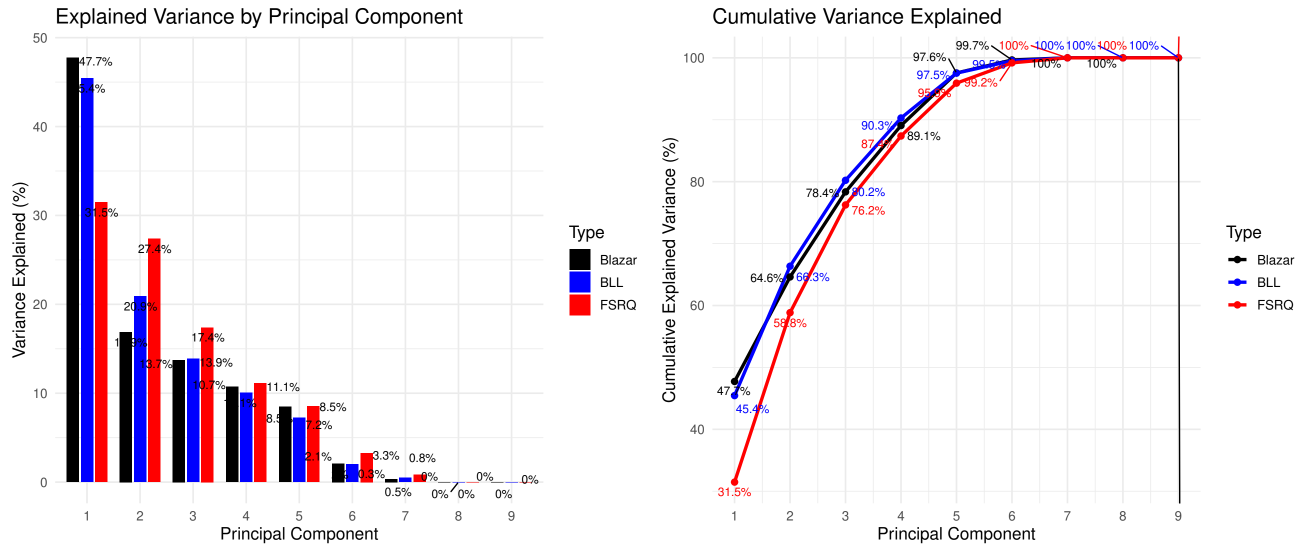

FIGURE 1 Variance explained by principal components for different blazar samples. Left: variance explained by the first nine principal components (PCs) for the entire blazar sample (black), the BL Lac subclass (blue), and the FSRQ subclass (red). Right: the corresponding cumulative variance curves. The FSRQ subclass shows a more complex multivariate structure, requiring more PCs to reach the same cumulative variance level as BL Lacs.

	Blazar			**BLL**			**FSRQ**		
Variable	PC_1	PC_2	PC_3	PC_1	PC_2	PC_3	PC_1	PC_2	PC_3
P_1	-0.72	0.01	0.41	-0.83	0.09	0.24	0.41	-0.21	0.80
P_2	-0.93	0.04	0.33	-0.96	0.10	0.20	0.57	-0.51	0.48
P_4	0.17	-0.17	0.12	0.23	-0.04	0.45	-0.09	-0.08	-0.03
P_5	-0.90	0.40	-0.09	-0.87	0.29	-0.33	0.96	0.22	-0.13
$\log L_p$	0.45	0.66	0.51	0.12	0.93	0.22	-0.32	0.74	0.08
$\log L_\gamma$	0.73	0.56	0.36	0.45	0.87	0.17	-0.22	0.89	0.37
$\Delta\nu$	-0.02	0.65	-0.73	0.36	0.25	-0.84	0.63	0.57	-0.46
CD	0.73	0.10	-0.07	0.74	0.12	-0.04	0.02	0.61	0.56
Γ_{PL}	0.90	-0.38	0.05	0.89	-0.28	0.25	-0.93	-0.24	0.09

TABLE 2 Variable loadings on the first three principal components for Blazar, BLL, and FSRQ samples.

Figure 3 shows the projection distribution of individual sources in the principal component space, revealing a continuous yet structured pattern of distribution for quasars across the first three principal component dimensions. Different source types are distinguished by color and shape. Overall, BLLs and FSRQs form a clear separation in the principal component space, with the most prominent distinction observed along the PC_1 direction. Along this axis, the mean values for BLLs and FSRQs are -1.554 and 1.851, respectively, with corresponding variances of 1.644 and 1.103, indicating a strong clustering tendency. This separation is highly consistent with previously identified classification indicators such as P_2, P_1, and $\Gamma_{\rm PL}$ (Abdo et al., 2010; Fossati et al., 1998; Ghisellini, Righi, Costamante, & Tavecchio, 2017).

In contrast, the separation is weaker along the PC_2 and PC_3 dimensions. For PC_2, the mean values for BLLs and FSRQs are 0.036 and 0.111, respectively, with variances of 1.272 and 1.266, suggesting similar distribution ranges along this direction. For PC_3, the means are -0.066 (BLL) and 0.139 (FSRQ), with variances of 1.174 (BLL) and 1.017 (FSRQ), indicating that FSRQs are slightly more concentrated in distribution. BCU sources exhibit some degree of overlap with both BLLs and FSRQs in the three-dimensional principal component space, suggesting that they may represent a mixture of the two classes or transitional types that are not yet clearly classified.

The structure of the principal component space reveals that the diversity of the quasar sample can be summarized into three

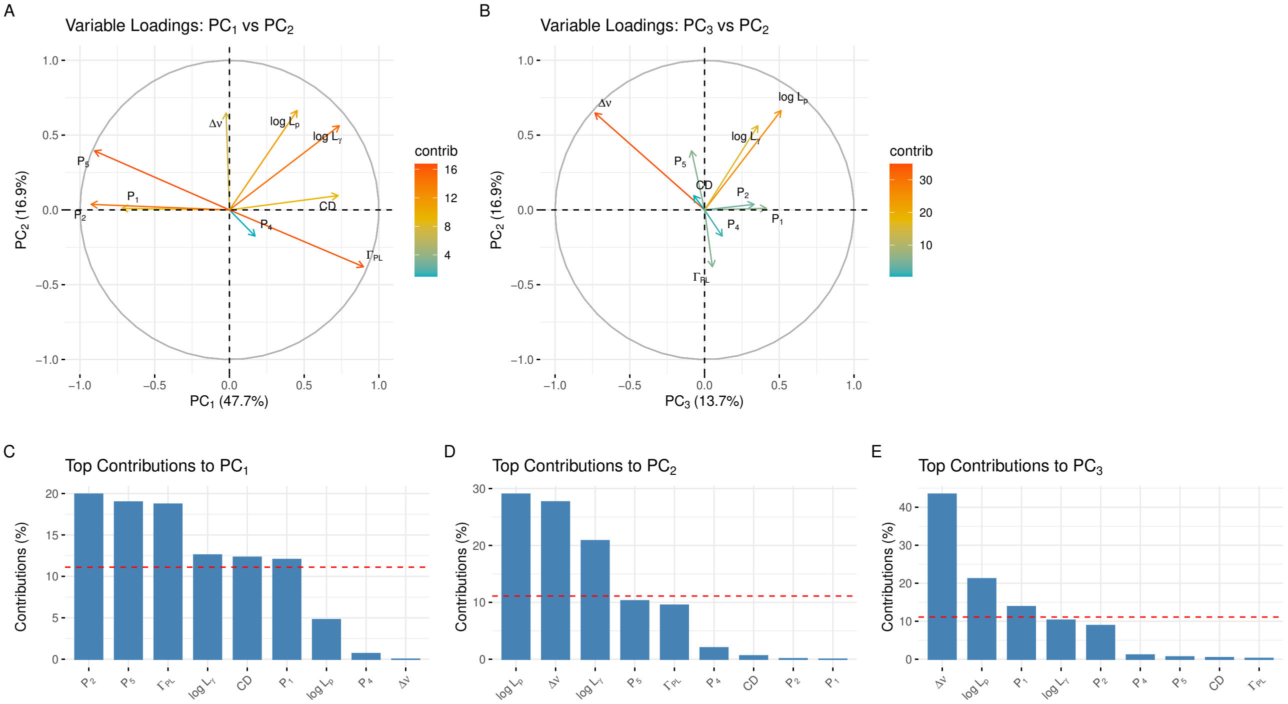

FIGURE 2 Relationships between the principal components and the original variables. (A) Variable loadings plotted on the $PC_1 - PC_2$ plane. (B) Variable loadings plotted on the $PC_3 - PC_2$ plane. (C-E) Top contributing variables to PC_1, PC_2, and PC_3, respectively. These plots highlight how different observed parameters contribute to the principal components, revealing distinct patterns that structure the variance in the blazar dataset.

major modes of variation: (1) spectral shape evolution across different bands (especially optical and high-energy bands); (2) variations in overall radiative brightness; and (3) the dominance of the synchrotron component in the energy spectrum. The clustering of BLLs and FSRQs in different regions of the principal component space implies fundamental differences in their physical driving mechanisms. In general, BLLs tend to exhibit higher synchrotron peak frequencies, steeper spectral curvature (P_1), and weaker high-energy spectral indices, whereas FSRQs lie on the other end of the energy spectrum, showing stronger high-energy components and different spectral shapes.

After performing PCA separately on the BLL and FSRQ subsamples, we found significant differences in their principal component structures. For the BLL subsample, the first three principal components explain 80.2% of the total variance. In contrast, the FSRQ subsample requires four components to reach 87.4% (see Figure 1). This suggests that FSRQs have higher parameter dimensionality and a more complex structural pattern.

In terms of explanatory variables, the leading features of the principal components differ between the two subsamples. For BLLs, the first principal component alone accounts for 45.4% of the variance, mainly driven by spectral shape parameters such as P_2, Γ_{PL}, P_5, P_1, and CD (see Table 2). For FSRQs, the first component explains only 31.5% of the variance and is primarily influenced by P_5, Γ_{PL}, $\Delta\nu$, and P_2. Although spectral shape remains dominant, the correlation among parameters is more dispersed.

A further comparison of the angles between the subsample and full-sample principal component axes (Figure 3) shows that BLLs are more aligned with the overall PCA structure. The angles between the BLL components and the full sample's PC axes are 16.8°, 26.7°, and 32.5° for PC_1, PC_2, and PC_3, respectively. In contrast, FSRQs show much larger angles: 144.6°, 34.5°, and 45.4°. This indicates that the principal component directions of FSRQs shift more significantly in parameter space.

These results support the physical distinction between BLLs and FSRQs and suggest that BLLs have a structure more consistent with the main features of the full quasar sample. We stress that the observed separation in the reduced parameter space does not imply a strict dichotomy between BLLs and FSRQs, but rather reflects dominant modes of variation in a continuous blazar population. The classification results should therefore be interpreted in a statistical and population-level context.

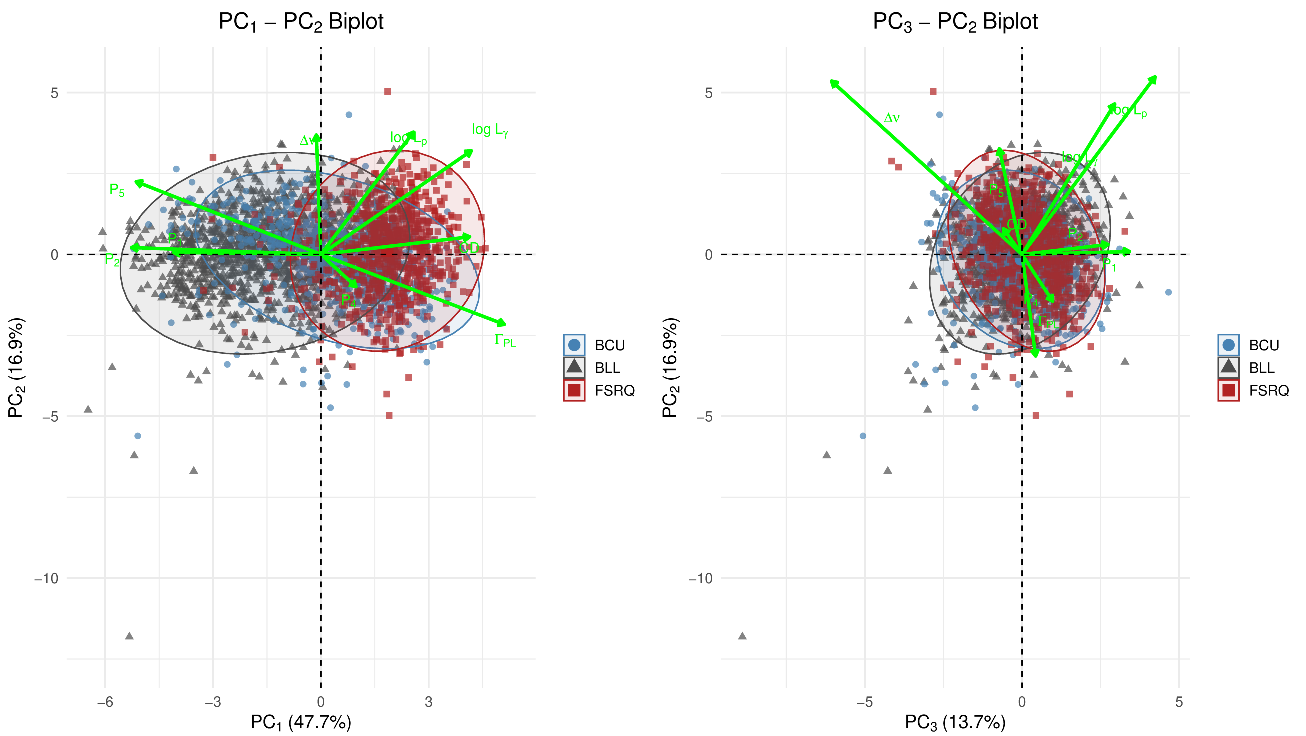

FIGURE 3 PCA biplots of the entire blazar sample projected onto two principal component planes. Panel A displays the distribution of individual sources on the $PC_1 - PC_2$ plane, and Panel B shows the distribution on the $PC_3 - PC_2$ plane. Points are colored by source type (BLL: blue, BCU: black, FSRQ: red), and ellipses represent 95% confidence regions for each type. The vectors indicate the directions and relative contributions of the original variables to the principal components. These projections illustrate the clustering and separation of different blazar types in the reduced parameter space.

3.2.2 | Classification Results (LDA)

We first apply the trained LDA model to the labeled sources (BLL and FSRQ) and analyze the distribution of their LD_1 scores Fisher (1936), as defined by Equation 5.

Figure 4 shows that BLLs tend to concentrate on the left side of the LD_1 axis, while FSRQs are located on the right, with a relatively narrow region of overlap. The mean and variance of the LD_1 values for BLLs and FSRQs are (-1.01, 1.12) and (1.52, 0.795), respectively.

This result indicates that LD_1—a multivariate axis primarily driven by P_1, P_4 and Γ_{PL}—offers strong separability between the two classes, consistent with the PCA-derived projection.

To evaluate the classification performance of LDA, we performed a 10-fold cross-validation using only the BLL and FSRQ training samples. The model achieved an overall accuracy of 90.6% (95% CI: 89.2%–91.9%) and a **Kappa coefficient** of $\kappa = 0.81$, indicating strong agreement between predicted and true labels beyond chance. Although LDA assumes linear class boundaries and similar covariance structures, the high cross-validated accuracy and Kappa coefficient indicate that these assumptions provide an adequate first-order approximation for the present dataset. Deviations from linearity are further examined using SVM as a complementary method.

The Cohen's Kappa statistic is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{6}$$

where p_o is the observed agreement (i.e., overall accuracy), and p_e is the agreement expected by random chance Cohen (1960).

The observed agreement is calculated as:

$$p_o = \frac{TP + TN}{N}, \tag{7}$$

where TP and TN represent the number of true positives and true negatives, and N is the total number of samples.

The expected agreement p_e is computed from the marginal totals of the confusion matrix:

$$p_e = \frac{(TP + FP)(TP + FN) + (FN + TN)(FP + TN)}{N^2}, \tag{8}$$

where $TP + FP$ and $FP + TN$ denote the predicted totals for positive and negative classes, respectively, and $TP + FN$ and $FN + TN$ are the corresponding actual totals.

The resulting value $\kappa = 0.81$ falls into the range of $0.8 \leq \kappa < 1.0$, which is generally interpreted as "almost perfect

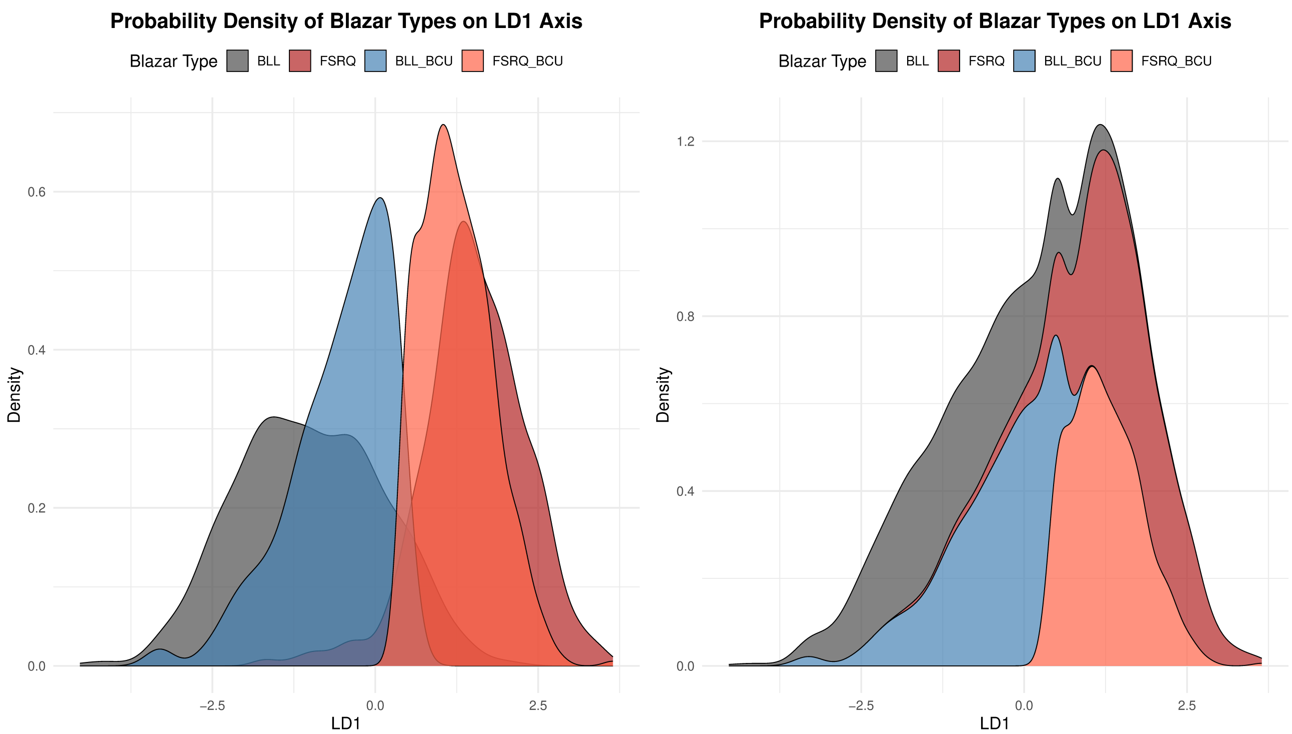

FIGURE 4 Probability density distribution of blazar sources along the first linear discriminant axis (LD_1) derived from LDA. The left panel shows the individual probability density functions (PDFs) for each blazar type, including known BLL and FSRQ sources, as well as BCU sources classified as BLL (BLL_BCU) or FSRQ (FSRQ_BCU). The right panel presents the same densities in a stacked format to illustrate the relative contribution of each type across the LD_1 axis. The clear separation between BLL and FSRQ populations supports the discriminative power of the LDA model in classifying BCU sources.

agreement." For reference, $\kappa = 1$ indicates perfect agreement, whereas values around 0.6 correspond to "substantial" agreement, and values below 0.4 indicate only "fair" or "poor" agreement Landis and Koch (1977).

In our experiment, the recall for BLLs was 88.5% (1011 out of 1142), and the specificity for FSRQs was 93.7% (711 out of 759). The positive predictive value (PPV) for BLL classification reached 95.5%, indicating a low false-positive rate. Although McNemar's test revealed a statistically significant asymmetry in misclassifications ($P < 0.001$) McNemar (1947), the practical impact was negligible, as reflected in the high Kappa and a balanced accuracy of 90.6%.

Subsequently, the LDA model is applied to the BCU sources. Their posterior probabilities with respect to the BLL and FSRQ classes are computed based on their LD_1 scores. As summarized in Table 4 , among the 807 BCU sources, we obtained probable classifications of 339 sources as BLLs and 468 as FSRQs. This posterior-based classification provides a statistically grounded basis for inferring the likely types of BCU sources.

3.2.3 | Classification Results Using Support Vector Machines (SVM)

To distinguish between BLLs and FSRQs, we applied an SVM classification approach using three different feature spaces. The dataset includes 1141 BLLs and 759 FSRQs. The feature spaces were constructed based on key physical parameters: synchrotron peak luminosity ($\log L_{syn}$), inverse Compton peak luminosity ($\log L_{\rm IC}$), and the Compton dominance parameter (CD). Specifically, we considered the following combinations: (1) $\log L_{syn}$ and CD, (2) $\log L_{\rm IC}$ and CD, and (3) a three-dimensional space comprising all three parameters. In all cases, the regularization parameter of the LinearSVC model was set to $C = 10^{-6}$. We deliberately adopt a linear SVM with strong regularization to prioritize model stability and interpretability over marginal gains in accuracy. Tests with non-linear kernels yield similar qualitative classification trends but introduce additional complexity and reduced transparency, which are not essential for the goals of this work.

The derived decision boundaries from these feature spaces were subsequently used to predict the classification of 807 BCUs. The classification results are summarized below:

1. **Feature space:** $\log L_{syn}$ **vs.** CD

- Decision boundary: $CD = -0.45 \log L_{syn} + 20.75$ (Fig. 5). This is equivalent to a classification axis defined by $SV_1 = CD + 0.45 \log L_p$.
- Correctly classified BLLs: 1003 out of 1142 (accuracy: 87.8%).
- Correctly classified FSRQs: 609 out of 759 (accuracy: 80.1%).
- Cohen's $\kappa = 0.683$.
- BCU predictions: 412 classified as BLLs, 395 as FSRQs.

2. **Feature space:** $\log L_{\rm IC}$ **vs.** CD

- Decision boundary: $CD = -0.49 \log L_{\rm IC} + 22.48$ (Fig. 5). This corresponds to a classification axis defined by $SV_2 = CD + 0.49 \log L_\gamma$.
- Correctly classified BLLs: 983 out of 1142 (accuracy: 86.2%).
- Correctly classified FSRQs: 616 out of 759 (accuracy: 81.2%).
- Cohen's $\kappa = 0.671$.
- BCU predictions: 383 classified as BLLs, 424 as FSRQs.

3. **Feature space:** $\log L_{syn}$, $\log L_{\rm IC}$, **and** CD

- Decision surface: $CD = -0.45 \log L_{syn} - 0.49 \log L_{\rm IC} + 43.1$ (Fig. 6). This corresponds to a classification axis defined by $SV_3 = CD + 0.45 \log L_p + 0.49 \log L_\gamma$.
- Correctly classified BLLs: 947 out of 1142 (accuracy: 83.0%).
- Correctly classified FSRQs: 628 out of 759 (accuracy: 82.7%).
- Cohen's $\kappa = 0.648$.
- BCU predictions: 382 classified as BLLs, 425 as FSRQs.

The classification performance across the different feature spaces indicates that two-dimensional spaces—particularly those involving $\log L_{syn}$ or $\log L_{\rm IC}$ with CD—achieve relatively higher classification accuracy for BLLs than for FSRQs. However, in the three-dimensional space, the classification accuracy for both classes becomes more balanced.

Applying the three decision boundaries to the 807 BCUs yielded consistent results across models: 349 sources were commonly predicted as BLLs and 425 as FSRQs. A small number of sources showed classification uncertainty, with 54 identified as possible BLLs (pBLLs) and 18 as possible FSRQs (pFSRQs), reflecting the limits of separation at the decision boundaries.

3.2.4 | Comparison of LDA and SVM Classification Results

The classification performance of LDA and SVM was evaluated using the labeled training set comprising 1,141 BLLs and 759 FSRQs. LDA achieved recall rates of 88.5% for BLLs and 93.7% for FSRQs, with a Cohen's Kappa coefficient of 0.81, indicating strong agreement and reliable class separation.

SVM performance varied with feature space selection. In the $\log L_{syn}$ vs. CD space, accuracies were 87.8% for BLLs and 80.1% for FSRQs, with $\kappa = 0.683$. In the $\log L_{\rm IC}$ vs. CD space, the respective accuracies were 86.2% and 81.2% ($\kappa = 0.671$). Using all three features ($\log L_{syn}$, $\log L_{\rm IC}$, and CD), performance slightly declined to 83.0% for BLLs and 82.7% for FSRQs ($\kappa = 0.648$). These results suggest that LDA offers more stable and consistently high performance, while SVM is more sensitive to feature selection.

Applying both classifiers to 807 BCUs yielded consistent results: 81.7% of sources were assigned to the same class by both methods. Specifically, 297 sources were identified as BLLs and 362 as FSRQs by both classifiers. The Cohen's Kappa coefficient was 0.63, indicating substantial agreement beyond chance. However, 148 sources showed inconsistent classifications—42 labeled as BLL by LDA but FSRQ by SVM, and 106 the reverse. These discrepancies likely reflect sources near the decision boundary and warrant further analysis.

4 | DISCUSSION

In this study, we constructed a high-dimensional parameter space for blazars using multiwavelength observational data, aiming to explore similarities and differences between blazar subclasses and to investigate their potential physical drivers. The parameters exhibit complex inter-correlations, reflecting the non-independent nature of the underlying physical quantities. Principal component analysis (PCA) indicates that a large fraction of the variance (78.4%) can be captured by the first three components, with the first component mainly related to luminosity-dependent parameters, the second dominated by spectral shape indicators, and the third associated with synchrotron dominance, suggesting that the data occupy a relatively low-dimensional subspace. These components are broadly associated with spectral shape, luminosity, and synchrotron dominance, although we caution that PCA identifies axes of maximal variance and does not necessarily imply direct causal relationships Fossati et al. (1998); Ghisellini et al. (2017); E. Massaro, Nesci, and Piranomonte (2012).

Building on this dimensionality reduction, we applied linear discriminant analysis (LDA) to identify axes that maximize

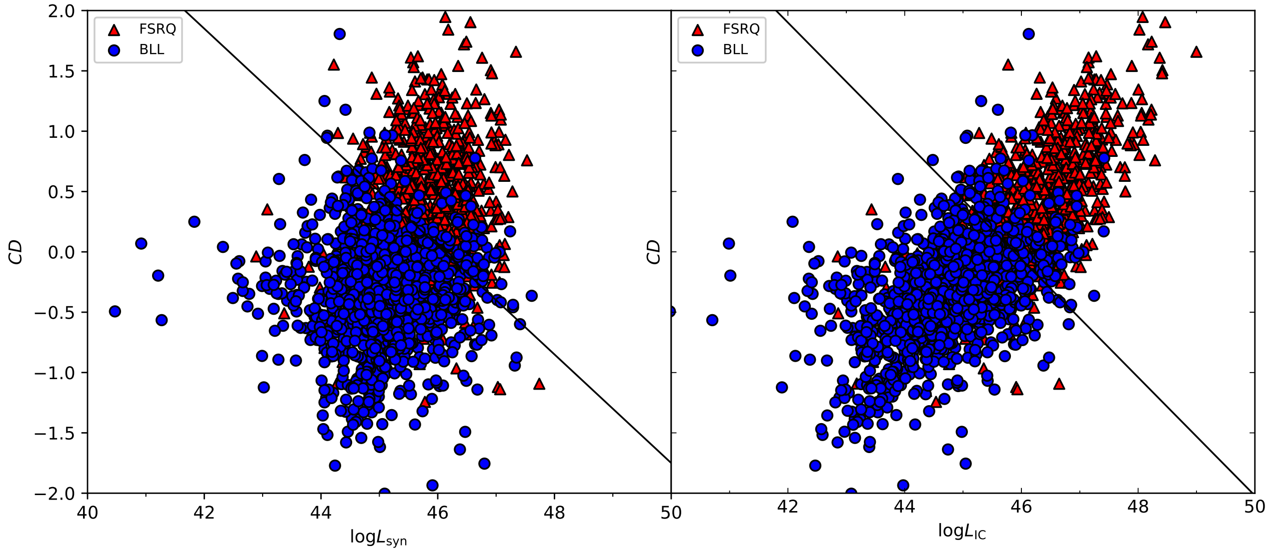

FIGURE 5 Compton dominance parameter (CD) plotted against synchrotron peak luminosity (log L_{syn}). Filled circles represent FSRQs, while filled triangles denote BLLs.

the separation between BLL and FSRQ sources. To further assess the robustness of the classification, we employed support vector machines (SVM) to determine optimal decision boundaries in the reduced parameter space Fan et al. (2022); Kang et al. (2019); Mao et al. (2016); Saz Parkinson et al. (2016); Van Soelen et al. (2017); Zhu et al. (2023). Both methods reveal broadly consistent separation trends, with spectral characteristics playing a major role in distinguishing the two subclasses. We also applied these approaches to reclassify BCUs, demonstrating the practical applicability of the method while emphasizing that classification reflects statistical patterns rather than definitive physical causation (Abdollahi et al., 2022).

The multi-parameter space of blazars is highly dimensional, with significant correlations between parameters. For example, the correlation coefficient between log L_p and log L_γ is $r = 0.851$, while Compton dominance (CD) shows weaker correlations of $r = 0.151$ and 0.647 with log L_p and log L_γ, respectively. These correlations suggest that blazar subclasses, such as BLLs and FSRQs, are not fully independent in terms of physical parameters, but rather exhibit continuous variation across multiple dimensions (Fossati et al., 1998; Ghisellini et al., 2017; E. Massaro et al., 2012). Consequently, the boundary separating subclasses may be complex and embedded within a high-dimensional structure, and parameter redundancy can lead to overlaps in low-dimensional projections, increasing classification uncertainty (Chiaro et al., 2016; Saz Parkinson et al., 2016).

We applied PCA to explore the underlying structure of the parameter space. The first three principal components, with PC_1 primarily associated with spectral shape, PC_2 dominated by luminosity-related parameters, and PC_3 reflecting synchrotron emission properties (mainly synchrotron peak frequency and/or Compton dominance), account for 78.4% of the total variance, indicating that a substantial portion of the sample variability can be represented in a reduced subspace. While these components capture statistical variance rather than causal physical effects, they provide a convenient framework for visualization and statistical classification, though the true class boundaries may still be complex and non-linear in the full parameter space (Fan et al., 2022; E. Massaro et al., 2012).

PCA indicates a statistically meaningful separation trend between BLLs and FSRQs, primarily along the first principal component (PC_1), which broadly reflects differences in spectral characteristics. Along the second and third components (PC_2 and PC_3), corresponding to luminosity and synchrotron dominance, the two classes show substantial overlap, highlighting the multi-dimensional nature of the parameter space. The parameters contributing most to PC_1—including synchrotron peak frequency (P_2), high-energy peak frequency (P_5), high-energy photon index (Γ_{PL}), gamma-ray luminosity (log L_γ), Compton dominance (CD), and synchrotron curvature (P_1)—capture the overall shape of the spectral energy distribution (SED) from optical to gamma-ray bands, consistent with the commonly observed trend that BLLs have harder, synchrotron-dominated spectra while FSRQs tend to show softer spectra influenced by external Compton processes (Abdo et al., 2010; Fossati et al., 1998; Ghisellini et al., 1998; Yang et al., 2026).

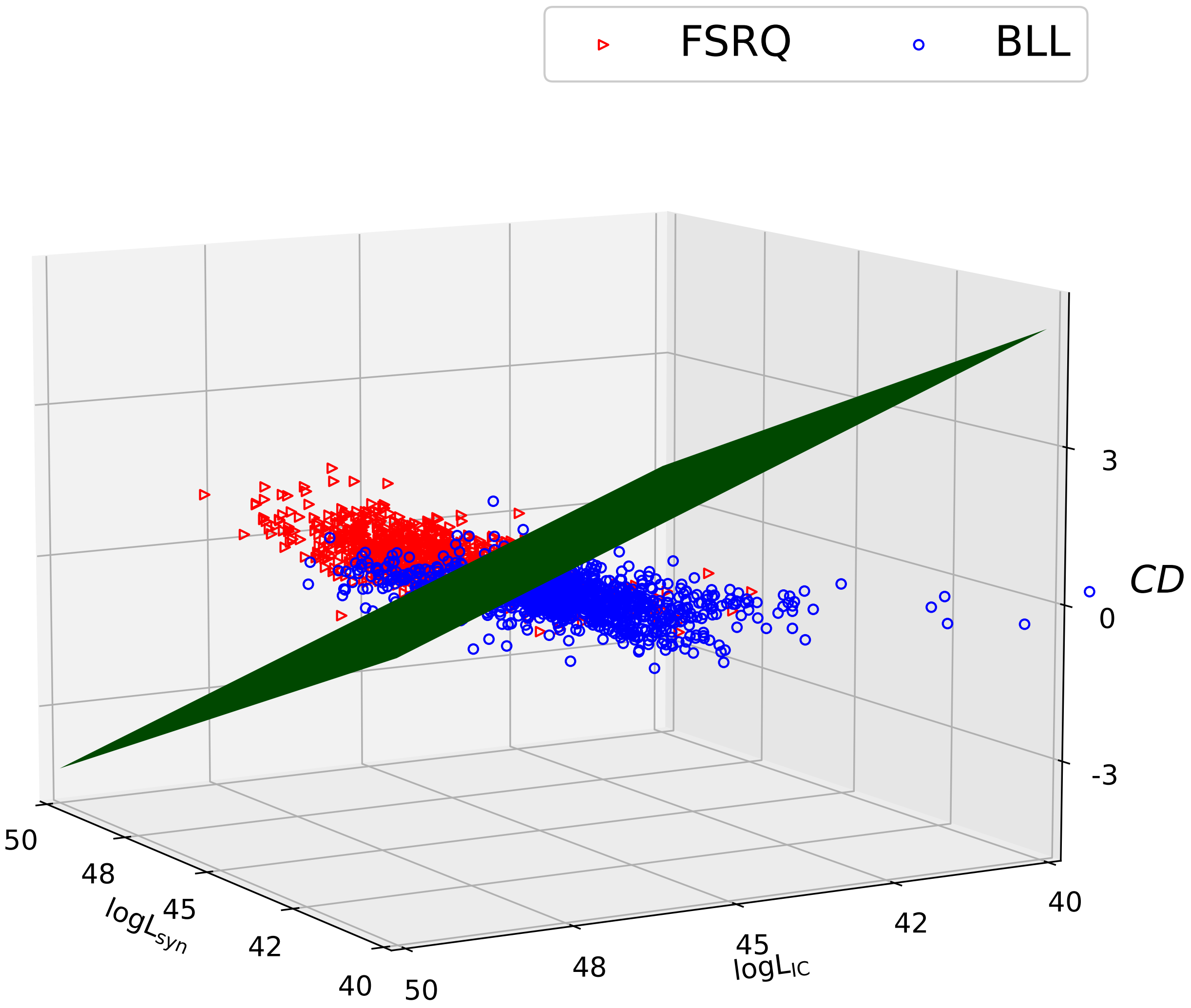

FIGURE 6 Compton dominance parameter (CD) plotted as a function of both synchrotron and Compton peak luminosities. Filled circles represent FSRQs, while filled triangles denote BLLs.

Linear discriminant analysis (LDA) supports this trend. The optimal discriminant axis (LD_1) effectively separates the two subclasses by construction, with the most influential variables largely overlapping those identified by PCA, emphasizing that spectral shape across multiple energy bands is central to blazar classification (Ghisellini et al., 2010; Mao et al., 2016). These results are consistent with a shared physical interpretation of spectral shape across multiple energy bands, while also acknowledging the statistical, rather than strictly causal, nature of the separation.

When analyzed separately, the two subclasses exhibit different effective dimensionalities. For BLLs, the first three principal components account for 80.2% of the variance, whereas for FSRQs this fraction decreases to 76.2%, indicating greater structural diversity in FSRQs. This difference may reflect additional physical complexity in FSRQs, such as stronger external photon fields and more prominent external Compton contributions, whereas BLLs display a more compact, lower-dimensional distribution consistent with a relatively homogeneous population dominated by synchrotron and synchrotron self-Compton processes (Ghisellini et al., 2010; Giommi et al., 2012; Padovani et al., 2017).

To assess the distribution characteristics and discriminative power of blazar classes in the physical parameter space, we

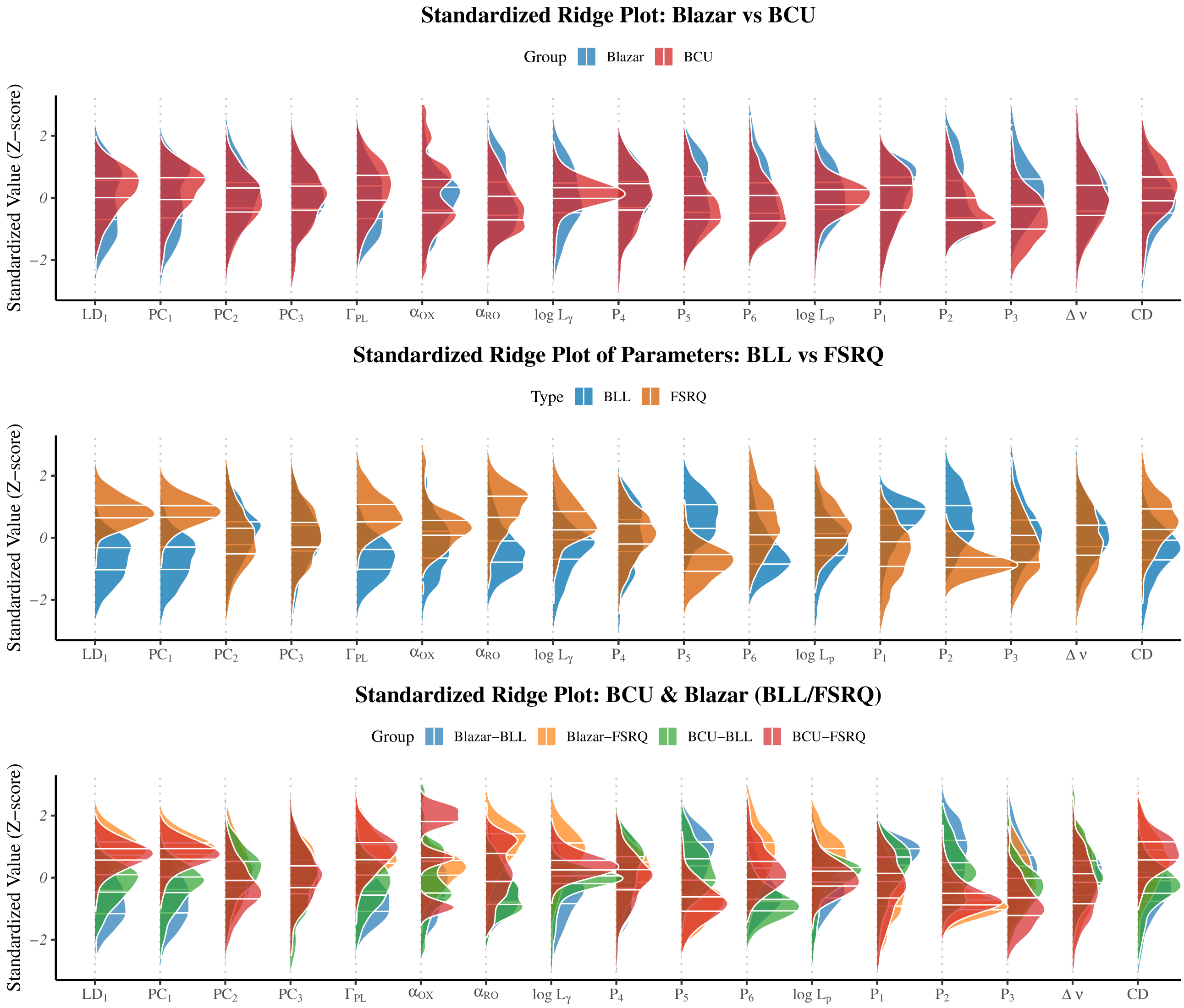

FIGURE 7 Comparison of parameter distributions for different types of blazars. Each ridge plot shows the distribution of blazars along 13 standardized parameters, including the physical discriminant LD_1, principal components PC_1 to PC_3, and nine other key parameters. All values are normalized (Z-score) to emphasize the shape of the distributions. Panel A groups the sources into two broad types: known blazars and BCUs. Panel B compares distributions between BLL and FSRQ types across all sources, including BCUs. Panel C separates blazars into four groups: known BLL and FSRQ sources, and BCU-classified BLL and FSRQ sources.

examined the distributions of BLLs, FSRQs, and BCUs across key parameters (Figure 7). The distributions show varying degrees of overlap, suggesting that some parameters alone may not provide sufficient separation for direct classification.

We quantified separability using the metric R (Section 3.1.1; Table 3). BLLs and FSRQs generally show good separability, with $R > 1$ in most parameter dimensions and an average $\bar{R} = 1.34$, consistent with their known physical differences. In contrast, BCUs overlap substantially with the known classes ($\bar{R} = 0.38$), particularly with FSRQs ($\bar{R} = 0.70$) and slightly less so with BLLs ($\bar{R} = 0.73$), highlighting the challenge of classifying these uncertain sources.

For model-predicted BCU subclasses, the separability from their corresponding known classes remains low, with BLL_BCU showing an average $R = 0.41$ relative to BLLs, and FSRQ_BCU an average $R = 0.52$ relative to FSRQs. These results indicate that the predicted BCU subclasses generally align with the physical characteristics of their known counterparts, though FSRQ_BCU exhibits greater variability, possibly reflecting intrinsic diversity or transitional sources.

In the PCA-projected space (Figure 1), BLLs and FSRQs occupy largely distinct regions, with PC_1 achieving $R = 2.58$, consistent with the interpretation that the joint multivariate parameter space captures the dominant class differences. Predicted BCU subclasses are also reasonably aligned with their respective known classes in this space, supporting the overall validity of our classification approach.

Overall, the combination of separability metrics and PCA projections provides a systematic evaluation of blazar subclass distinguishability. The analysis reinforces the clear separation between BLLs and FSRQs while illustrating the inherent complexity in classifying BCUs, offering a quantitative foundation for future classification efforts based on physical parameters.

The complementarity of LDA and SVM highlights both the interpretability and robustness of blazar classification. LDA identifies a linear discriminant axis in the full parameter space, emphasizing physically meaningful differences between BLLs and FSRQs. SVM, applied to a three-parameter subspace (log $L_{\rm p}$, log L_γ, and Compton dominance CD), yields a consistent decision boundary with LDA, despite slightly lower overall accuracy (83

The parameters used in SVM exhibit some correlations, which can introduce nonlinear effects in low-dimensional projections and lead to uncertainty for borderline cases. In contrast, LDA maintains a global linear discriminant, offering clearer interpretability and deeper physical insight. Together, these methods show that combining statistical and machine learning approaches can effectively capture the underlying structure of blazar populations while providing reliable guidance for subclass assignments.

Our results are broadly consistent with previous classification studies of blazars, while providing a more comprehensive and physically grounded framework. Earlier works often relied on single parameters, such as synchrotron peak frequency or spectral index, or on two-dimensional empirical cuts. For instance, classifications based on synchrotron peak and radio luminosity captured systematic differences in SED shapes, while the combined use of high-energy photon index and luminosity improved accuracy. Our PCA and LDA analyses recover these trends in a high-dimensional parameter space, with spectral shape and Γ_{PL} emerging as dominant axes, supporting their physical relevance.

Compared with infrared or X-ray color-based methods, our approach generalizes the classification to a multidimensional space, allowing for systematic and interpretable separation of BLLs and FSRQs. The use of SVM further validates the robustness of the decision boundaries identified by LDA, showing that combining dimensionality reduction with machine learning enables both accurate classification and insight into the underlying physical distinctions. Overall, this framework extends traditional low-dimensional methods, offering a more unified and extensible approach to blazar subclassification.

To assess the consistency of our classification with previous studies, we compared the results for the 807 BCU sources in our sample with several recent works. Across the major studies (Agarwal, 2023; Fan et al., 2022; Kang et al., 2019; P.-Y. Xiao et al., 2023; Yang et al., 2022), the LDA classifier achieves agreement rates ranging from 83% to 91%, with Cohen's Kappa values between 0.667 and 0.812, indicating moderate-to-substantial consistency. The SVM classifier shows slightly lower agreement, typically between 78% and 82%, reflecting its sensitivity to parameter correlations and borderline cases. These comparisons confirm that both LDA and SVM recover the dominant physical distinctions between BLLs and FSRQs identified in previous works, while offering a high-dimensional, physically interpretable classification framework.

For subsets of sources previously proposed as blazar candidates (Yang et al., 2022), our classifications achieve 73% consistency with the latest 4FGL-DR4 catalog (Ballet et al., 2023), further supporting the reliability of our approach. Notably, our PCA+LDA+SVM methodology not only aligns with prior empirical or low-dimensional classifications, but also provides a unified, statistically grounded basis for future blazar subclass identification. **As a preliminary independent validation, we cross-matched the 807 BCUs classified in this work with the SDSS spectroscopic sample analysed in the work (G. Chen et al., 2026), obtained 44 commnon sources, of which 41 have consistent classifications, corresponding to an agreement rate of $\sim$ 93.2% (see Table 4). The three discrepant sources, 4FGL J0040.9+3203, 4FGL J0147.6-0825, and 4FGL J1243.0+3950, have spectral signal-to-noise ratios (S/N) of 1.19, 3.50, and 5.29, respectively, all lower than the mean value of 11.21 for the 84 spectra analysed by G. Chen et al. (2026). The classification discrepancies were therefore likely caused by the low S/N of these spectra. Further spectroscopic validation is nevertheless required. We are currently carrying out follow-up observations of selected BCUs with the 2.4-m LiJiang Telescope (LJT; Wang et al., 2019) and have also cross-matched our BCU sample with spectroscopic data from the Dark Energy Spectroscopic Instrument (DESI; DESI Collaboration et al., 2025). These efforts will provide further tests of the predicted classifications and help determine the redshifts and other physical parameters (e.g. black hole mass, Eddington luminosity, and accretion properties) for the uncertain sources.**

Analysis of the physical parameter distributions for BCU sources indicates that, for certain parameters such as P_3, BCUs tend to exhibit systematically lower values, which may reflect

TABLE 3 Separability Scores $R(A, B)$ of Key Parameters Between Blazar Source Types

Parameter	R(BLL, FSRQ)	R(BCU, Known)	R(BLL, BCU)	R(FSRQ, BCU)	$R(\mathrm{BLL_{BCU}}, \mathrm{BLL})$	$R(\mathrm{FSRQ_{BCU}}, \mathrm{FSRQ})$
LD_1	2.72	0.47	1.36	1.10	0.64	0.56
P_2	2.71	0.38	1.10	1.13	0.38	0.38
PC_1	2.58	0.42	1.24	1.06	0.46	0.48
L_{BRL}	2.53	0.40	1.60	0.85	0.92	0.71
P_5	2.24	0.37	1.19	0.79	0.34	0.17
Γ_{PL}	2.17	0.42	1.26	0.67	0.36	0.20
$\log L_R$	1.97	0.54	0.93	1.21	0.52	0.98
α_{RO}	1.75	0.34	0.14	1.63	0.23	0.96
SV_1	1.47	0.47	0.96	0.66	0.43	0.53
SV_3	1.47	0.52	0.92	0.75	0.48	0.67
SV_2	1.45	0.42	0.97	0.58	0.40	0.42
$\log L_\gamma$	1.42	0.58	0.84	0.85	0.51	0.84
P_1	1.20	0.11	0.59	0.54	0.11	0.19
z	1.19	0.09	0.63	0.46	0.15	0.11
CD	1.18	0.39	0.88	0.28	0.29	0.23
P_6	1.14	0.39	0.08	1.09	0.77	0.67
$\log L_p$	1.02	0.35	0.46	0.83	0.35	0.93
$\log L_O$	0.86	0.54	0.74	0.39	0.65	0.38
α_{OX}	0.65	0.24	0.36	0.45	0.30	0.47
$\log L_X$	0.47	0.24	0.17	0.49	0.26	0.58
P_3	0.46	0.97	1.16	0.70	0.87	0.95
P_4	0.32	0.09	0.10	0.24	0.07	0.22
PC_3	0.20	0.06	0.07	0.18	0.09	0.17
$\Delta\nu$	0.18	0.10	0.16	0.07	0.19	0.23
PC_2	0.08	0.20	0.18	0.23	0.28	0.56
Mean	1.34	0.38	0.73	0.70	0.41	0.52

Note. Each $R(A, B)$ denotes a separability measure between classes A and B. The final row lists column-wise means. Values rounded to two decimal places.

their low-luminosity nature or the impact of observational limitations. For other key parameters, including α_{OX} and Γ_{PL}, BCUs largely overlap with known BLL and FSRQ sources, suggesting that classification uncertainties arise not solely from intrinsic physical differences but also from observational selection effects Ackermann et al. (2015); Chiaro et al. (2016). Nevertheless, statistical distinctions remain between the combined BCU sample and the known blazar population, indicating that currently classified sources may not fully represent the overall population.

Following the reclassification of BCUs into BLL_BCU and FSRQ_BCU using LDA and SVM, we find that BLL_BCU sources are largely indistinguishable from the original BLLs across all examined parameters, with separability ratios consistently below 1. In contrast, FSRQ_BCU sources show greater separation from the original FSRQs in several parameters, suggesting a higher diversity within this subclass. Overall, the parameter characteristics of BCUs are more similar to those of FSRQs, consistent with their observed spectral properties. These results demonstrate the feasibility and physical validity of using the LDA discriminant axis LD_1 for systematic reclassification of uncertain sources, reinforcing previous machine-learning-based efforts to classify BCUs (L. Chen, 2018; Chiaro et al., 2016; Fan et al., 2022; Kang et al., 2019).

Among the 807 BCU sources analyzed, 148 exhibit inconsistent classifications between LDA and SVM. These sources have scores near zero on the LDA discriminant axis LD_1, placing them close to the classification boundary and suggesting that their physical properties occupy a transitional or ambiguous region between BLLs and FSRQs (Ghisellini et al., 2017; Ghisellini et al., 2011; Giommi et al., 2012). Projection of these sources into common parameter subspaces, such as the $\log L_p$-CD plane, shows a relatively scattered distribution rather than clustering near the SVM decision boundary, indicating that the separation between BLLs and FSRQs is likely governed by a complex, high-dimensional surface rather than a simple linear hyperplane.

These ambiguous sources may therefore reflect a physical continuity between BLL and FSRQ subclasses or represent objects undergoing spectral evolution, corresponding to potential transitional states of high scientific interest (Ghisellini et al., 2017; Mishra et al., 2021). Their uncertain classification underscores the limitations of low-dimensional projections and highlights the need for targeted follow-up studies. Future multi-wavelength monitoring, long-term spectral analyses, and investigations into "changing-look" behaviors will be particularly valuable for elucidating the nature and evolution of these sources.

Despite the advances achieved in this study, several methodological limitations warrant consideration. First, some physical parameters, such as the synchrotron peak frequency, are derived from model fitting (e.g., parabolic or log-parabolic forms), which can introduce systematic uncertainties that propagate into the principal components and classification axes (Massaro et al., 2004; Nieppola, Tornikoski, & Valtaoja, 2006). Second, our classification relies primarily on linear techniques such as LDA and SVM. While these methods provide strong interpretability and clear insights into physical distinctions between blazar subclasses, they are inherently limited in capturing complex nonlinear boundaries, particularly for ambiguous or transitional sources (Hastie, Tibshirani, & Friedman, 2009).

Future work can address these limitations through several complementary strategies. Integrating the interpretability of LDA with the flexibility of SVM, or employing ensemble learning methods such as Bagging and Boosting, may improve classification robustness and adaptability (Hastie et al., 2009). Nonlinear algorithms, including Random Forests, XGBoost, or neural networks, offer additional potential for modeling the high-dimensional, complex boundary morphology between BLL and FSRQ sources (Breiman, 2001; Friedman, 2001). Expanding the parameter space to include multi-band variability, polarization measurements, and time-lag features could further enhance the characterization of underlying physical processes, while incorporating long-baseline, multi-wavelength temporal data would provide dynamic insights complementary to static parameter-based classification, particularly for identifying "changing-look" or transitional sources (Ghisellini et al., 2017).

Finally, careful attention to overfitting and model validation remains essential. The regularization parameter in SVM, for example, must balance bias and variance to ensure that high training performance generalizes to unseen data (Cortes & Vapnik, 1995a; Schölkopf & Smola, 2001). Our successful application of SVM, combined with its agreement with existing BCU classifications, demonstrates that when properly regularized, the method can achieve reliable and consistent performance across diverse datasets (Vapnik, 2000). Overall, by refining these approaches and incorporating additional physical and temporal features, future studies can establish a more nuanced and robust framework for blazar classification while preserving the interpretability of the model.

5 | CONCLUSION

In this study, we applied a combination of statistical and machine learning techniques to classify blazars and investigate the physical distinctions between BL Lac objects (BLLs) and flat-spectrum radio quasars (FSRQs). Principal component analysis (PCA) revealed that most of the variance in blazar properties is captured by three key factors: spectral shape, luminosity, and synchrotron dominance. The first principal component, dominated by spectral features, shows a strong correlation with the observed separation between BLLs and FSRQs, highlighting the intrinsic physical differences between these subclasses. BLLs display a more compact, low-dimensional distribution consistent with synchrotron-dominated emission, whereas FSRQs show higher dimensional complexity, likely reflecting additional processes such as external Compton scattering.

Building on this structure, linear discriminant analysis (LDA) identified a physically interpretable axis that maximally separates the two classes, while a support vector machine (SVM) provided a complementary decision boundary capable of resolving ambiguous cases. Applying these methods to 807 BCUs yielded broadly consistent classifications, with discrepancies concentrated near the class boundary, suggesting potentially ambiguous classifications or sources with intermediate properties; however, such *transitional* states cannot be uniquely identified from static parameters alone.

Together, the PCA-LDA-SVM framework demonstrates both robust classification performance and interpretability in high-dimensional parameter space, supporting the interpretation that the BLL-FSRQ distinction reflects genuine, physically meaningful differences. This approach provides a unified, statistically grounded methodology for reclassifying uncertain blazars and offers a foundation for future studies exploring blazar diversity, evolution, and transitional phenomena.

ACKNOWLEDGMENTS

We would like to thank the anonymous reviewer for the comments and suggestions! This work is partially supported by the National Natural Science Foundation of China (grant Nos.12433004, 12133004, 12303019), National Key Research and Development Program of China, Grant No. 2025YFA1614102, the National SKA Program of China (grant No. 2022SKA0120102, No. 2025SKA0130100), the Eighteenth Regular Meeting Exchange Project of The Scientific and Technological Cooperation Committee between the People's Republic of China and the Republic of Bulgaria (Series No. 1802), the Guangdong Basic and Applied Basic Research Foundation (grant Nos. 2023A1515012529, 2024A1515012218), the Innovation Research Team of Guangzhou University (grant No. 2023ZDP001), the Natural Science Foundation of Hunan Province (grant No. 2024JJ5067), and the Hunan Provincial Department of Education (grant No. 22B0438). This research was partially supported by the Bulgarian National Science Fund of the Ministry of Education and Science under grants KP-06-H68/4 (2022), KP-06-H88/4 (2024) and KP-06-KITAJ/12

TABLE 4 Classification results for 807 BCUs

4FGL name	Class[1]	$logL^1_{\nu_p}$	$logL^2_{\nu_{IC}}$	$logCD$	C_{LDA}	C_{m1}	C_{m2}	C_{m3}	C_{TW}	C_F	C_K	C_A	C_C
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
4FGL J0001.2+4741	ISP	45.05	45.06	0.01	BLL	BLL	BLL	BLL	BLL	BLL	BLL	ambiguous	-
4FGL J0002.4-5156	HSP	45.44	44.91	-0.53	BLL	BLL	BLL	BLL	BLL	BLL	NN	BLL	-
4FGL J0003.1-5248	HSP	45.50	45.46	-0.04	BLL	BLL	BLL	BLL	BLL	BLL	BLL	BLL	-
4FGL J0003.3-1928	LSP	46.21	46.42	0.21	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	BLL	FSRQ	ambiguous	-
4FGL J0008.0-3937	LSP	45.05	45.86	0.81	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	NN	-
4FGL J0008.4+1455	HSP	45.10	45.50	0.40	BLL	FSRQ	FSRQ	FSRQ	FSRQ	BLL	BLL	BLL	-
4FGL J0009.8-6358	LSP	44.83	46.61	1.78	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	NN	BLL	-
4FGL J0010.8-2154	LSP	45.30	45.17	-0.13	FSRQ	BLL	BLL	BLL	BLL	BLL	FSRQ	ambiguous	-
4FGL J0011.4-4110	LSP	44.94	45.38	0.44	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	NN	ambiguous	-
4FGL J0012.0+7043	LSP	45.32	45.90	0.58	FSRQ	FSRQ	FSRQ	FSRQ	FSRQ	BLL	FSRQ	ambiguous	-
…(Table continues for all 807 sources)…													

[1] For a work by J. H. Yang et al. (2022); [2] For a work by J. Yang et al. (2023); C_{m1}: Classification from $logL_{\nu_{syn}}$ and CD; C_{m2}: Classification from $logL_{IC}$ and CD; C_{m3}: Classification from $logL_{syn}$, $logL_{IC}$ and CD; C_F: Classification in the work by Fan et al. (2022); C_K: Classification in the work by Kang et al. (2019); C_A: Classification in the work by Agarwal (2023); C_C: Classification in the work by G. Chen et al. (2026).This table is available in its entirety in machine-readable forms.

(2024). We also thank Chongjing Li for his help with the data compilation.

Author contributions

W.-X. Yang and J.-H. Fan designed the study. W.-X. Yang, G.-H. Chen, J.-T. Zhu, H.-G. Wang, D. Bastieri, X.-H. Ye, J.-C. Liang, and D.-X. Wu performed data collection and analysis. All authors contributed to the interpretation of results and manuscript preparation.

Financial disclosure

None reported.

Conflict of interest

The authors declare no potential conflict of interests.

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